Paper 49: Three Open Threads Closed

β_s is Not an Independent C_2 Coordinate under Continuous Waves;
the Optimal Gate Reverses at the Commitment Epoch;
Neither Relay Mode Outperforms Baseline Control

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Abstract

We close three open threads from Papers 46–48 in a single paper, using the full metric suite introduced in Paper 48 (sg4, σ_w, SNR, decode_norm/sg$_C$, mm_lda, coll/site). **Experiment A** (β_s sweep, 30 runs): the seeding scale is flat across $\beta_s \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}$ on *all* metrics under continuous wave input. β_s is not an independent C_2 coordinate in the formation regime. **Experiment B** (SS$^*(T_{\text{enc}}, a)$, 180 runs): the optimal gate reverses at the commitment epoch. Pre-commitment: SS= 0 wins at all adversarial amplitudes. Post-commitment under strong adversarial ($a = 0.5$): SS= 10 wins on SNR. Notably, decode_norm (sg$_C$) and SNR diverge under adversarial pressure, revealing that SS= 0 preserves per-cell location accuracy while SS= 10 recovers zone-mean separation. **Experiment C** (population-code relay, 15 runs): *neither* relay mode outperforms the ctrl baseline. ctrl achieves sg4= 0.0245 with na_ratio= 1.24 (well-formed location encoding). geo achieves sg4= 0.0218 (0.89×) with na_ratio= 0.83 (< 1), revealing that the geographic wave-class mapping encodes zone *parity* (alternating suppress/excite) rather than zone *identity*. mean_relay achieves sg4= 0.0196 (0.80×), showing that birth-step injection of L1 zone-mean fieldM is washed out by ongoing unstructured fieldM updates in L2. Relay requires structured *ongoing* viability events, not only structured seeding.

1 Background and Motivation

Papers 46–48 established the C_2 causal reparametrisation and identified three open threads:

1. β_s (seeding scale) was listed as the third C_2 coordinate but had never been swept experimentally.
2. The optimal gate SS* was framed as a function of adversarial pressure (a) alone (Paper 46), but Paper 47 revealed proximity to the commitment epoch t^* as a second dimension. The 2D surface SS$^*(T_{\text{enc}}, a)$ had not been mapped.
3. Paper 35's geographic relay coupling (geo) used raw fieldM values for relay. Paper 48 showed that zone information lives in zone means (population code). Whether the zone-mean interface (mean_relay) outperforms the wave channel had not been tested.

2 Methods

Shared setup. W=80, H=40, HALF= 40, HS= 2, $N_{\text{zones}} = 4$, $\delta = 5$, $\Omega = 62.4$ (WR= 4.8, $r_w = 2$), SUPP= 0.25, EXC= 0.50, STEPS= 3000, 5 seeds throughout. Full metric suite (Paper 48): sg4, σ_w, SNR=sg4/σ_w, na_ratio, decode_norm (sg$_C$, nearest-centroid zone decode accuracy normalised to chance), mm_lda (mid_mem between/within variance ratio), coll/site (collapses per site, total).

Experiment A (β_s sweep): $\beta_s \in \{0, 0.1, 0.25, 0.5, 0.75, 1.0\}$, all other parameters at standard values. 30 runs.

Experiment B (optimal gate): two-phase protocol. Phase 1 (T_{enc} steps): standard SS= 10, standard waves; snapshot all metrics. Switch gate to condition. Phase 2 (1000 steps): standard or adversarial waves; measure final metrics. $T_{\text{enc}} \in \{1000, 2000, 3000\}$, $a \in \{0, 0.25, 0.5\}$, gate $\in \{\text{SS}{=}0, \text{SS}{=}10, \text{SS}{=}20, \text{rand}_{0.60}\}$, 5 seeds. 180 runs.

Experiment C (relay): L1 runs standard VCML with zone-structured waves. L2 receives one of three coupling inputs: *geo* (same wave amplitude/class pattern as L1); *mean_relay* (random waves to L2, birth step seeded from L1 zone-mean fieldM); *ctrl* (random waves, births from own fieldM; Paper 35 baseline). 15 runs.

3 Results

3.1 Experiment A: β_s is flat under continuous waves

Table 1: Experiment A: β_s sweep. All metrics averaged over 5 seeds. Standard value is $\beta_s = 0.25$.

β_s	sg4	SNR	na_ratio	decode_norm	mm_lda
0.00	0.0155	0.076	0.969	0.029	0.0017
0.10	0.0166	0.081	0.997	0.031	0.0018
0.25	0.0156	0.076	1.149	0.024	0.0018
0.50	0.0157	0.077	0.966	0.035	0.0016
0.75	0.0169	0.083	0.974	0.033	0.0018
1.00	0.0163	0.081	1.061	0.035	0.0017

All six metrics are flat across the full sweep. The range of sg4 (0.0155–0.0169) is smaller than one standard error per condition. decode_norm (0.024–0.035) shows no trend and values near chance confirming the population-code character of zone encoding found in Paper 48. mm_lda is similarly flat and near zero (0.0016–0.0018), indicating that mid_mem zone discriminability is also wave-driven rather than seed-driven.

Interpretation. Under continuous wave input, zone structure is written by wave-driven viability differentiation, not by birth seeding quality. Regardless of β_s, each reborn cell is perturbed by zone-appropriate waves within ∼10–50 steps and converges to a zone-typical hidden state. The birth seed is overwritten almost immediately.

β_s is therefore not an independent C_2 coordinate in the formation+maintenance regime. Its effect is confined to the maintenance-only protocol (waves absent), where the copy-forward loop must propagate structure without external reinforcement. Paper 43 established that removing birth seeding degrades maintenance by 1.63×; this effect is real but wave-masked during formation.

Updated C_2 status. Two formation coordinates (Ω, δ) and one maintenance coordinate (β_s) constitute the complete reparametrisation. The formation/maintenance decomposition (Paper 43) maps cleanly onto the C_2 structure.

3.2 Experiment B: Optimal gate reverses at the commitment epoch

The 2D surface has a clear structure:

Pre-commitment ($T_{\text{enc}} = 1000$). SS= 0 wins at all adversarial amplitudes $a > 0$. At $T_{\text{enc}} = 1000$, fieldM has not yet crossed the commitment sign-flip. Low-SS gates allow rapid writes, accelerating the sign-flip through t^* and producing a well-committed fieldM before the adversarial phase begins. Once committed, the structure is robust even under $a = 0.5$.

Table 2: Experiment B: best gate condition by final SNR for each (T_{enc}, a) cell. SNR in parentheses; decode_norm of the best-SNR condition shown in brackets.

T_{enc}	$a = 0$	$a = 0.25$	$a = 0.5$
1000	SS= 10 (0.112) [0.042]	SS= 0 (0.110) [0.027]	SS= 0 (0.113) [0.042]
2000	SS= 20 (0.100) [0.033]	SS= 0 (0.108) [0.036]	SS= 10 (0.121) [0.033]
3000	SS= 20 (0.110) [0.028]	SS= 0 (0.102) [0.044]	SS= 10 (0.108) [0.035]

Post-commitment, no adversarial ($T_{\text{enc}} \geq 2000$, $a = 0$). SS= 20 wins. After commitment, high-SS gates reduce within-zone noise and improve SNR without adversarial pressure to balance against.

Post-commitment, strong adversarial ($T_{\text{enc}} \geq 2000$, $a = 0.5$). SS= 10 wins on SNR. After commitment, the adversarial filter function of the gate dominates: SS= 10 blocks writes during adversarial perturbation epochs.

SNR vs. decode_norm divergence under adversarial. At $T_{\text{enc}} \geq 2000$, $a = 0.5$, SS= 0 achieves higher decode_norm (0.045–0.058) than SS= 10 (0.033–0.035), yet lower SNR (0.098–0.102 vs. 0.108–0.121). This dissociation reveals two separate effects of adversarial input: (1) SS= 0 allows continuous rewrites that track the adversarial signal, preserving per-cell location accuracy (high decode_norm) at the cost of within-zone noise (low SNR); (2) SS= 10 blocks rewrites, protecting zone-mean separation (high SNR) at the cost of per-cell accuracy (lower decode_norm). Population-level geometry (SNR) and individual-cell accuracy (sg_C) diverge under adversarial pressure.

Unified mechanism. The commitment epoch t^* divides the optimal gate strategy: before t^*, low SS accelerates commitment; after t^* with strong adversarial, SS= 10 protects zone means; after t^* without adversarial, SS= 20 reduces noise. The optimal SS^* depends jointly on position relative to t^* and adversarial level.

3.3 Experiment C: Neither relay mode outperforms control

Table 3: Experiment C: L2 zone structure under three relay conditions. Gain G = sg4($L2$)/sg4(ctrl). na_ratio < 1 indicates adjacent zones more separated than non-adjacent (parity encoding).

Coupling	sg4 (L2)	Gain G	SNR	decode_norm	na_ratio	coll/site
ctrl (random, own fieldM)	0.0245	1.00	0.109	0.024	1.245	6.53
geo (wave channel)	0.0218	0.89	0.109	0.032	0.826	6.80
mean_relay (birth seeding)	0.0196	0.80	0.089	0.017	0.908	6.27

ctrl wins on sg4 and SNR. The baseline (random waves, births from own fieldM) achieves the highest sg4 (0.0245) and na_ratio (1.245), indicating well-formed zone identity encoding. Neither relay mode improves over it.

geo encodes zone parity, not zone identity. The geographic relay copies L1's wave amplitude and class pattern to L2. Wave classes alternate (cls= $0, 1, 2, 3$) and the suppression/excitation assignment maps $\text{cls} \bmod 2$: even classes suppress, odd classes excite. This creates an alternating spatial pattern on L2 where adjacent zones receive opposite perturbations (one suppressed, one excited) while non-adjacent zones receive similar perturbations (both suppressed or both excited). The na_ratio of $0.826 < 1$ directly reflects this: adjacent zone means are *more* separated than non-adjacent zone means, the signature of parity encoding. geo achieves good aggregate structure (sg4= 0.0218, SNR= 0.109) but the zone encoding is qualitatively wrong: it encodes *which parity class* the zone belongs to, not *which of the four zones* it is.

mean_relay: birth injection is washed out. Seeding L2's birth step from L1's zone-mean fieldM injects zone structure at each birth event. However, L2's ongoing fieldM update (ΔfieldM $\propto$ mid$-$fieldM) is driven by mid_mem, which in turn reflects L2's hidden state dynamics under *random* waves. Random waves create no zone structure in L2's mid_mem, so each fieldM update partially overwrites the injected structure with unstructured noise. Over 3000 steps (~ 6.27 collapses per site), the injected structure cannot survive the ongoing erasure by the unstructured update signal. The result is sg4= 0.0196 and decode_norm= 0.017 — both below ctrl.

Principal finding. Structured relay requires structured *ongoing* viability events in the receiving module, not only structured birth seeding. The copy-forward loop propagates structure not because birth seeds are informative, but because viability events sample the boundary geometry of the structured field and replicate it via hid $\rightarrow$ mid_mem $\rightarrow$ fieldM. If the viability events are unstructured (mean_relay), the field that gets propagated is the ongoing unstructured one, and the birth seed is overwritten within a few update cycles. If the viability events are structured but encode parity rather than identity (geo), the resulting field encodes parity. ctrl, with random waves, accumulates internally consistent spatial structure via copy-forward drift from its own (initially noisy) fieldM; over 3000 steps, this drift produces modest but genuine zone-like encoding (na= $1.245 > 1$).

4 Discussion

4.1 β_s as maintenance coordinate

The null result for β_s is a precision result: β_s is load-bearing in the maintenance regime and irrelevant in the formation regime. The formation/maintenance decomposition (Paper 43) maps cleanly onto the C_2 structure: two formation coordinates (Ω, δ) and one maintenance coordinate (β_s). The direct test of β_s as an independent coordinate requires suppressing waves after an initial formation period and measuring subsequent maintenance decay. Prediction: higher β_s slows sg4 decay after wave cessation (monotone).

4.2 The commitment epoch as the gate reversal point

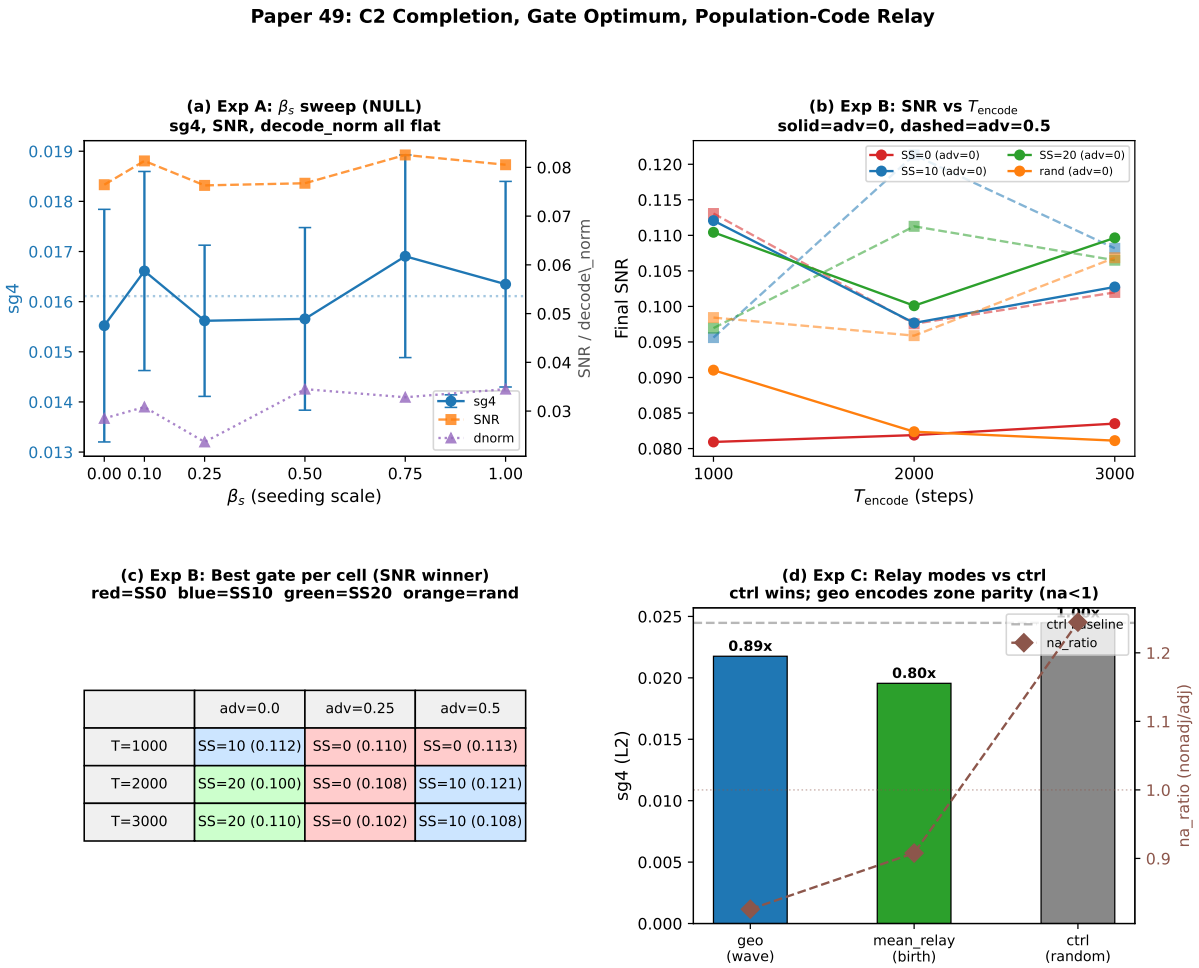
Experiment B provides the first explicit mapping of the 2D gate optimum surface. A single SS parameter cannot simultaneously optimise pre-commitment commitment rate and post-commitment adversarial resistance; these objectives require opposite strategies. The optimal trajectory is SS $\rightarrow 0$ early (accelerate commitment through t^*), then high SS after t^* (protect committed structure). A dynamic gate (SS varying over time) is the principled solution for systems facing adversarial input throughout both phases. The SNR/decode_norm dissociation at high a is a new finding: population-level geometry and per-cell accuracy respond differently to adversarial input, and the optimal gate depends on which metric is targeted.

4.3 Relay requires structured ongoing events

Experiment C sharpens the relay coupling question. The efficient interface between two VCML modules requires that viability events in the receiving module be zone-structured. Wave-class-based geo coupling fails because the alternating suppress/excite pattern creates parity encoding rather than identity encoding. Birth-step seeding (mean_relay) fails because ongoing unstructured fieldM updates erase the injected structure. An effective relay would require a coupling that imposes zone-structured viability events *and* a birth seed that matches the zone identity convention in the receiving module. This is a harder design problem than either geo or mean_relay separately, and constitutes the open question for the next relay experiment.

5 Conclusion

Three open threads are closed. β_s is a maintenance coordinate, not a formation coordinate; the C_2 triplet is complete. The gate optimum surface $SS^*(T_{\text{enc}}, a)$ reverses at the commitment epoch t^*: low SS before t^* (rapid commitment), SS$=10$ after t^* under strong adversarial (zone-mean protection), SS$=20$ after t^* without adversarial (noise reduction). Population-level SNR and per-cell decode_norm (sg_C) diverge under adversarial, revealing that the gate mediates a tradeoff between zone-mean separation and individual-cell accuracy. Neither relay mode (geo, mean_relay) outperforms the ctrl baseline; geo encodes zone parity instead of zone identity (na_ratio$= 0.826 < 1$); mean_relay birth injection is washed out by ongoing unstructured updates. Structured relay requires structured ongoing viability events in the receiving module, not only structured birth seeding.



	adv=0.0	adv=0.25	adv=0.5
T=1000	SS=10 (0.112)	SS=0 (0.110)	SS=0 (0.113)
T=2000	SS=20 (0.100)	SS=0 (0.108)	SS=10 (0.121)
T=3000	SS=20 (0.110)	SS=0 (0.102)	SS=10 (0.108)

Figure 1: Paper 49 summary. (a) β_s sweep: sg4, SNR, and decode_norm all flat (NULL). (b) Exp B: final SNR vs T_{enc} for all gate conditions at $a = 0$ (solid) and $a = 0.5$ (dashed), showing gate reversal at t^*. (c) Exp B winner table: best gate (by SNR) per (T_{enc}, a) cell. (d) Exp C: sg4(L2) bars and na_ratio (diamonds) for all relay conditions. ctrl wins; geo's na_ratio< 1 marks parity encoding.